

AI and the Language of DNA/RNA

Midsem Project Plan: Work Division for Tasks 1–3

Executive Summary

This document details the work division and project roadmap for our AI team (Omkar, Arya, Vinayak, Parth, and Sahil). For the upcoming midsem presentation, we are targeting the completion of the first three core objectives defined in our proposal:

- **Task 1:** Representing DNA/RNA as a language (k-mers).
- **Task 2:** Exploring the sequence language via statistics and visualization.
- **Task 3:** Applying Natural Language Processing (NLP) ideas to genomic data.

CONCURRENT EXECUTION STRATEGY

A common pitfall in data processing pipelines is the bottleneck created by sequential dependencies—where the entire team waits for the dataset to be cleaned and formatted before beginning their respective tasks. To ensure maximum productivity leading up to the midsem presentation, we will utilize a **mock-data driven parallel execution strategy**.

Step 1: The API Contract (Day 1)

The team will agree on the exact format of the intermediate data file. It will be a standard tabular format (e.g., CSV) containing an ID, the raw sequence, a class label (disease/non-disease), and a string of space-separated k-mers.

Step 2: Mock Generation

Omkar will spend the first 30 minutes generating a dummy dataset containing 50 randomized strings of 'A', 'T', 'G', 'C' formatted to this exact schema, mimicking the final pipeline output.

Step 3: Parallel Development

Arya, Vinayak, Parth, and Sahil will use this dummy CSV to build, test, and debug their statistical scripts, NLP models, and visualization tools immediately.

Step 4: Hot Swap Integration

Once Omkar completes the rigorous extraction of real biological sequences, he will replace the dummy file. The team's scripts will run on the real data with zero architectural changes.

MIDSEM PRESENTATION ROADMAP

The timeline below is designed to guide the team over a six-week period leading to the midsem review. It balances foundational engineering with the exploratory and NLP analyses required for the presentation.

Phase	Timeline	Key Objectives & Deliverables
Phase 1: Foundation	Weeks 1–2	• Day 1: Agree on data schema; Omkar distributes mock dataset. • Week 1: Omkar secures biological database access and begins curation. • Week 2: Omkar finishes the k-mer extraction pipeline (Task 1). Arya begins computing baseline metrics on the newly extracted real data.
Phase 2: Exploration	Weeks 3–4	• Week 3: Arya completes word frequency analysis and identifies rare/common biological sequence patterns (Task 2). • Week 4: Vinayak generates the first PCA and t-SNE dimensionality reduction plots to observe class clustering (Task 2).
Phase 3: NLP Integration	Weeks 4–5	• Week 4: Parth establishes the Bag-of-Words and TF-IDF baseline matrices (Task 3). • Week 5: Sahil finalizes word2vec embeddings and contextual sequence representations (Task 3).
Phase 4: Synthesis	Week 6	• Cross-evaluate TF-IDF representations against Word2Vec contexts. • Compile the curated dataset pipeline and visualizations into slide format. • Dry-run the Midsem Presentation focusing on insights generated from Tasks 1-3.

Midsem Deliverables Target

For the presentation, we must deliver: 1) A clear demonstration of the sequence-to-language representation pipeline. 2) Visualizations of the learned sequence space. 3) An analytical comparison of how different classical and advanced NLP representations perform when structuring biological "language".

DETAILED WORK DIVISION: ENGINEERING & STATISTICS

Omkar

Task 1: Pipeline & Data Representation

Primary Responsibility: Building the foundational bridge between raw biological data and machine learning frameworks. Omkar is the owner of the data pipeline.

- **Mock Data Generation (Day 1):** Create the initial dummy dataset to unblock the rest of the team. Establish the exact formatting schema (e.g., ID, class label, space-delimited k-mers).
- **Data Curation:** Source raw DNA/RNA sequences from public biological repositories. Handle noise in the data (e.g., removing invalid sequence characters or non-canonical bases).
- **Algorithmic Tokenization:** Treat A, T/U, G, and C strictly as an alphabet. Develop a robust, sliding-window algorithm to convert contiguous sequences into "words" using k-mers.
- **Parameterization:** Ensure the pipeline can easily toggle between different values of k (e.g., 3-mers, 4-mers, 6-mers) so the downstream team can evaluate how token size impacts information capture.
- **Output Handoff:** Export the cleaned, tokenized datasets (treating sequences effectively as tokenized paragraphs) to local storage for the NLP and visualization tools.

Arya

Task 1 & 2: Sequence Statistics

Primary Responsibility: Quantitative exploration of the language. Arya bridges the tokenized data output with preliminary statistical insights before complex NLP is applied.

- **Frequency Analytics:** Ingest Omkar's tokenized dataset and compute term frequencies for the genomic "words".
- **Pattern Identification:** Identify the most frequently occurring k-mers (akin to "stop words" in English) and highly rare k-mers. Observe if certain words only appear in disease-causing sequences.
- **K-mer Evaluation:** Analyze the combinatorial explosion. Document how moving from 3-mers (64 possible combinations) to 6-mers (4096 combinations) affects sparsity and frequency distributions.
- **Baseline Statistics:** Generate sequence statistics such as average length, vocabulary richness, and class distribution, preparing statistical summaries for the midsem presentation slides.

DETAILED WORK DIVISION: VISUALIZATION & NLP

Vinayak

Task 2: Visualization of Sequence Space

Primary Responsibility: Visual storytelling of the data. Vinayak must determine if treating DNA as a language naturally reveals hidden structures through visual clustering.

- **Dimensionality Reduction:** Apply algorithms like PCA (Principal Component Analysis), t-SNE, and UMAP to the numerical representations generated by the team.
 - **Class Clustering:** Map disease vs. non-disease classes onto 2D and 3D scatter plots to identify if the sequence representations naturally group together without explicit model training.
 - **Dashboarding / Presentation Graphics:** Clean and format these plots for the midsem presentation, ensuring axes are clearly defined and the clustering phenomena (or lack thereof) are easily explainable to the audience.
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Parth

Task 3: Classical NLP Approaches

Primary Responsibility: Applying foundational natural language processing heuristics. Parth treats the sequences as plain-text documents.

- **Bag-of-Words (BoW):** Utilize libraries (e.g., scikit-learn's `CountVectorizer`) to represent whole DNA/RNA sequences based purely on k-mer presence and count.
 - **TF-IDF Implementation:** Calculate the Term Frequency-Inverse Document Frequency for the sequences. Determine if down-weighting highly common biological k-mers improves the clarity of the sequence vectors.
 - **Vector Handoff:** Provide these sparse matrix representations to Vinayak for baseline visualization.
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Sahil

Task 3: Advanced NLP & Embeddings

Primary Responsibility: Semantic modeling. Sahil explores whether the language of DNA possesses "context" similar to how semantic meaning is captured in human language.

- **Word2Vec/Embeddings:** Train embedding models (like Gensim's Word2Vec) on the tokenized sequences to map k-mers into dense vector spaces.
- **Contextual Similarity:** Investigate whether sequence words that frequently occur in similar biological contexts acquire mathematical similarity in their vector representations.
- **Sequence-Level Embeddings:** Explore techniques (e.g., Doc2Vec or average word embeddings) to represent an entire DNA/RNA sequence as a single dense vector, comparing this continuous representation against Parth's sparse TF-IDF vectors.